

7th CII 3M Compétition 2020

A Kaizen Presentation By

everest

EVEREST INDUSTRIES LIMITED
(Podanur Works – Coimbatore)

Company Profile

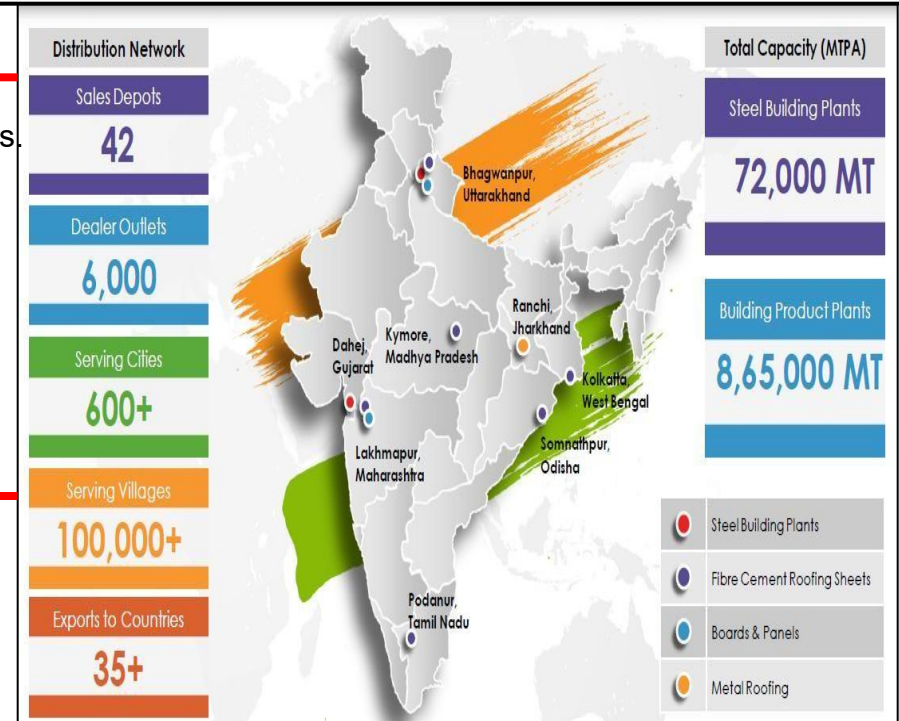
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Company Overview:

Everest Industries Limited, incorporated in 1934, has a rich history in manufacturing of Building products and Steel products. Everest offers a complete range of roofing ceiling, wall, flooring and cladding products distributed through a large network, and also pre-engineered steel buildings for industrial, commercial and residential applications. It is one of the leading building solution providers in India, providing detailed technical assistance in the form of design, drawings and implementation for every project.

Our Business:

- **Building Products (63%)** – include fiber cement roofing sheets, fiber cement boards, solid wall panels.
- **Steel buildings (37%)** – offers customized building solutions like Pre-Engineered Steel buildings, metal roofing sheets and cladding.



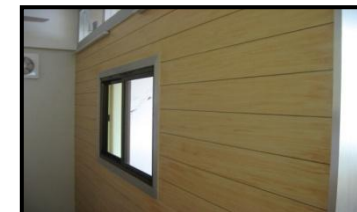
Fibre Cement Roofing Sheets



Hi-Tech Roofing Sheets



Super Sheets



Cement Planks



Fibre Cement Boards



Rapicon Walls



Smart Steel Buildings



Our Esteemed Customers

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Kaizen Theme: To eliminate Machine stoppages for Cutters & Shoes change.

Implemented Area: Line No -1, Podanur Works, Coimbatore

Implementation Date: 04th November 2019

Cross Functional Team:



Mr. M. Moorthy
Prod. Shift Incharge



Team Leader
Mr. Y. Kesavaraj
(PW1 Line Incharge)



Mr. N. Anil Kumar
(Engineer - QS & BE)



Mr. M. Madan Raj
Prod. Shift Incharge



Mr. N. Karthikeyan
(Prod. Shift Incharge)



Mr. N. Karthikeyan
(Engineer - Production)



Mr. Jagir
(Fitter)

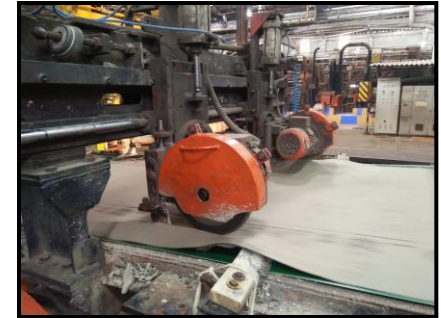
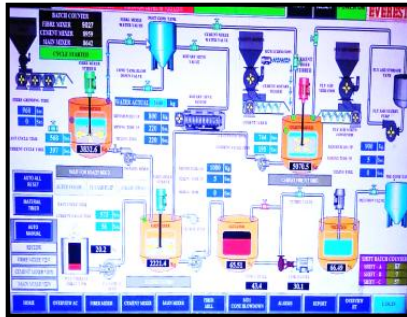


Mr. Edward
(Fitter)

Problem / Present Status

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Process Flow:



**RAW MATERIAL
PREPARATION**

VATS

SHEET FORMING

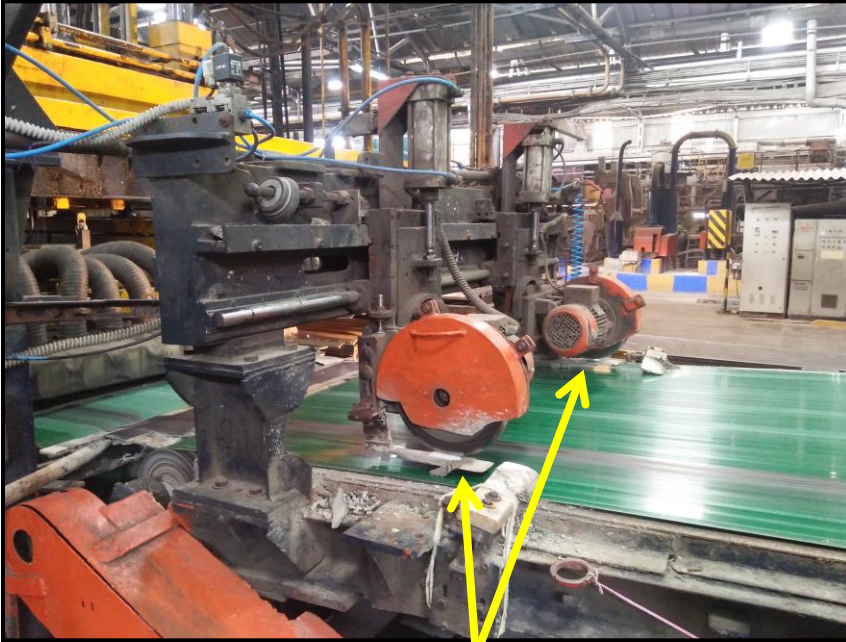
SHEET CUTTING

DE-PILING

PRE CURING

**SHEET HANDLING /
PILING MACHINE**





Cutters Unit



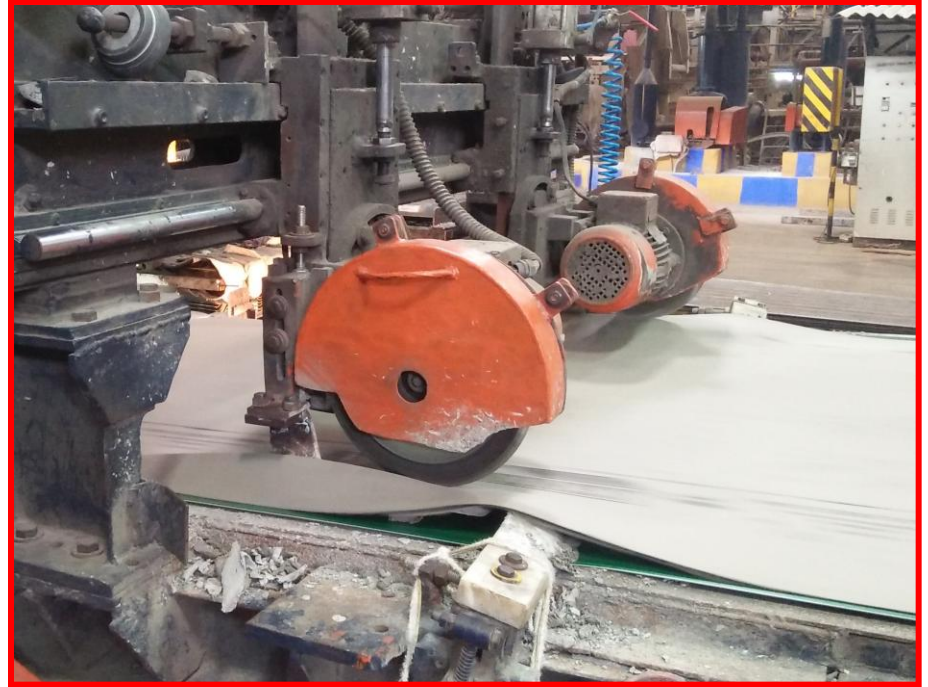
Fitter Changing Cutters with Spanner when Machine is stopped

Problem/ Present Status :

Loss of Production i.e., 3.54 M.T per day due to Machine Stoppage for 15 minutes daily for cutters change.



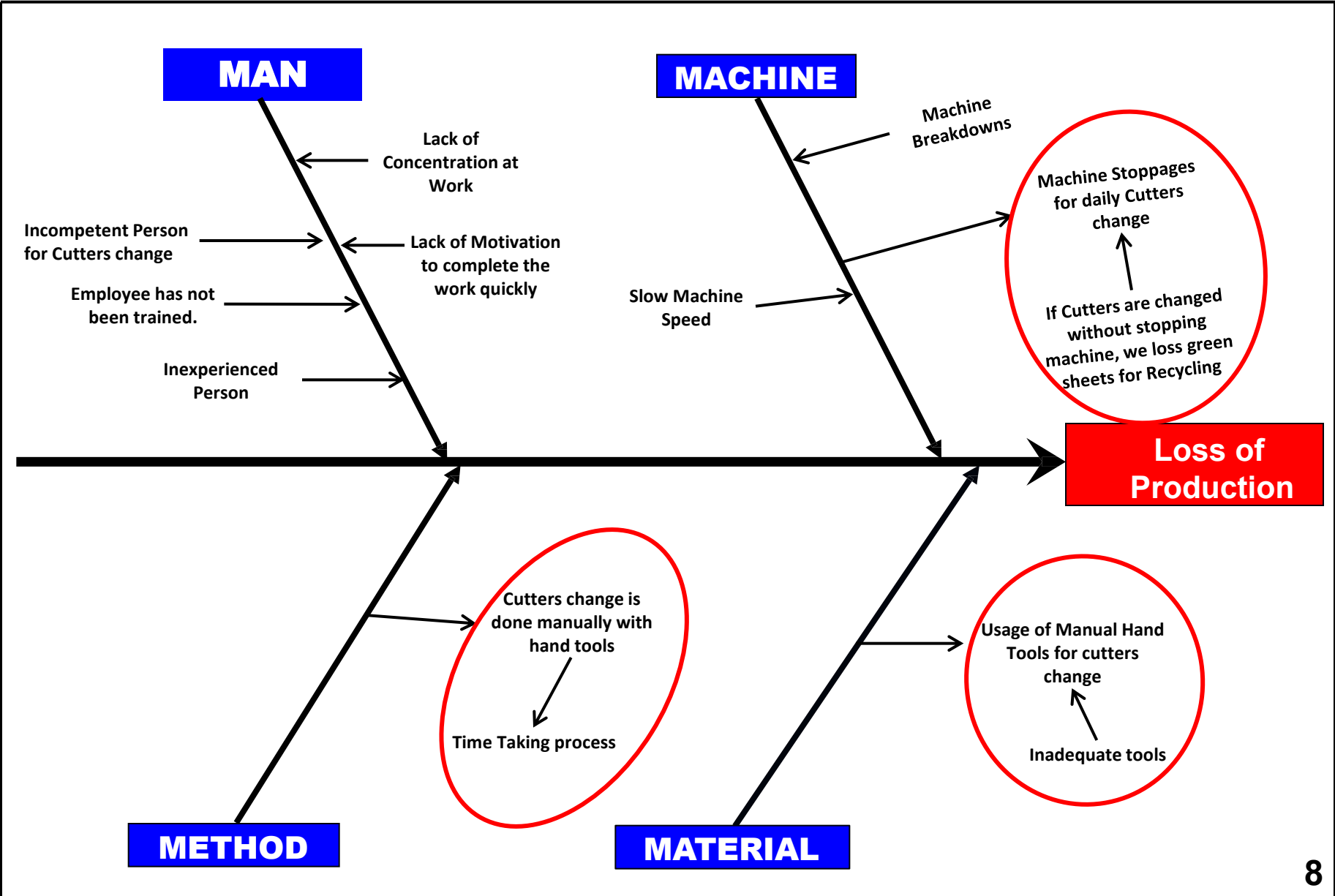
TCT Cutter



TCT Cutter Unit

- Due to abrasive raw materials, while cutting TCT cutters sharpness will reduce. Due to this sheet cutting finishing is not good in our product. For this we have to change TCT cutters for every 24 Hours.

Cause & Effect Analysis



Problem Statement	Loss of Production
Why 1	Due to Machine Stoppage.
Why 2	Machine Stoppages for Cutters & Shoes Change.
Why 3	If Machine not stopped, we loss wet green sheets for recycling.
Why 4	There is NO process to collect Green wet sheets while changing cutters & shoes to eliminate sheet wastage.

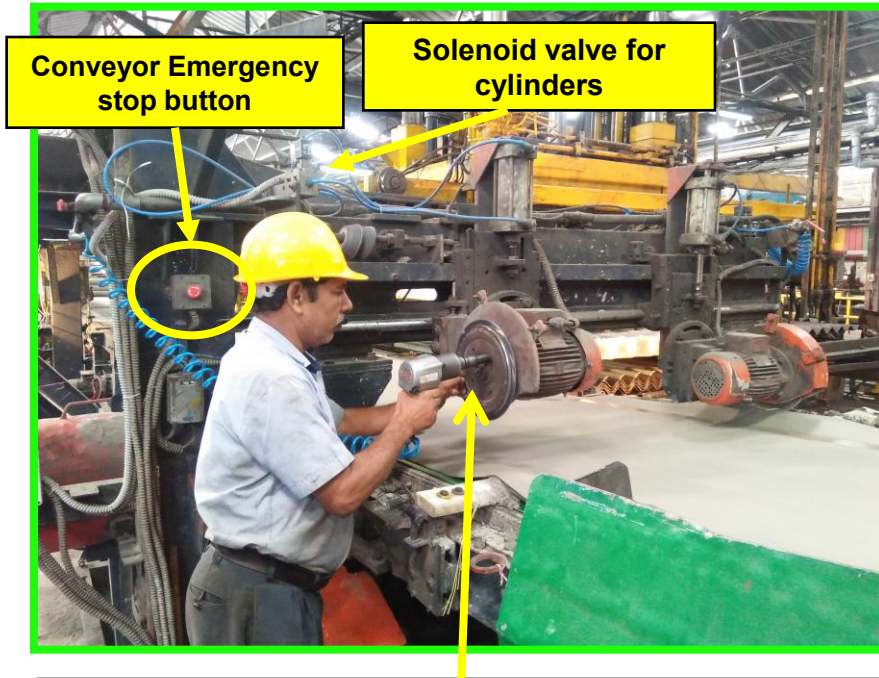
Root Cause: There is NO Process to collect Green sheets for moulding during Cutters change.

Idea Generation: To change Cutters & Shoes while giving sheets for moulding & to use Pneumatic gun to quickly change cutters to eliminate Machine Stoppages



Action: To collect sheets for moulding during cutter change to eliminate machine stoppages & to use pneumatic gun for changing cutters

SN	Counter Measure Activities	Start Date	Target Date	Responsibility	Status
01	To provide individual solenoid valves for each cutter Pneumatic cylinder Up Down movement.	25/10/2019	04/11/2019	Maintenance & Production Department	Completed
02	To procure & use Pneumatic Gun for Screwing & Unscrewing of bolts used for fixing cutters.	25/10/2019	04/11/2019	Production Line Incharge	Completed
03	To collect wet green sheets for Hand Moulding goods while changing cutters.	04/11/2019	04/11/2019	Production Department	Completed
04	To install Conveyor Emergency stop button near cutter unit.	02/11/2019	02/11/2019	Electrical Department	Completed



Fitter Changing Cutters with Pneumatic Gun



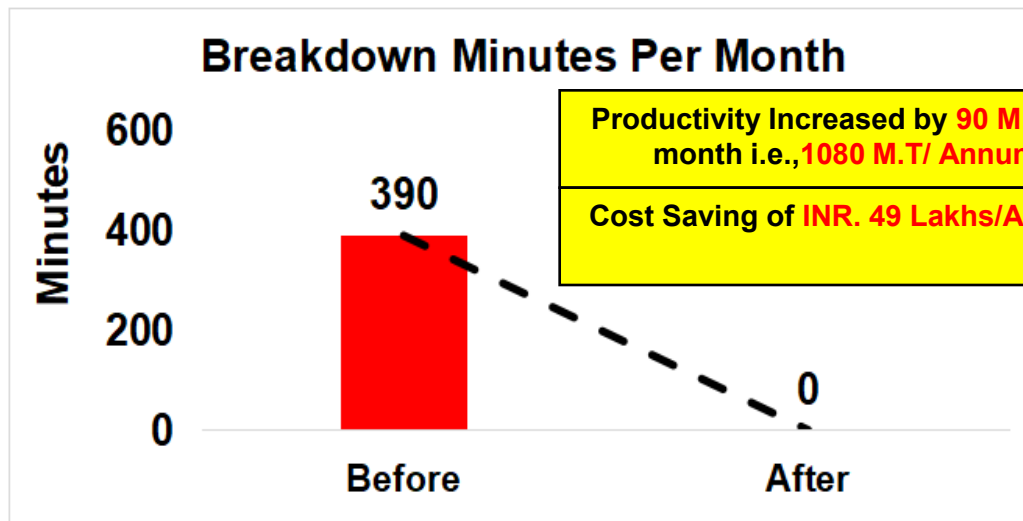
Sheets Collecting for Moulding

- During Sheet collecting for moulding, we are making each cutter ideal one by one & simultaneously changing cutters accordingly.



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Before		After	
Machine Down Time Per Day	15 Min	Machine Down Time Per Day	NIL
Machine Down Time Per Month	390 Min	Machine Down Time Per Month	NIL
Production Loss Per Day	3.54 M.Tons	Production Loss Per Day	NIL
Production Loss Per Month	90 M.Tons	Production Loss Per Month	NIL
Reduction in Cost of Production Per Month (Conversion & Overhead Cost)	No Profit	Reduction in Cost of Production Per Month (Conversion & Overhead cost)	Rs. 1.26 Lakhs
Reduction in Cost Production Per Year (Conversion & Overhead Cost)	No Profit	Reduction in Cost of Production Per Year (Conversion & Overhead cost)	Rs. 15.2 Lakhs



Validation of Kaizen in 7 Wastes of Manufacturing

T	Transportation	NA
I	Inventory	NA
M	Motion	Corrected
W	Waiting	Corrected
O	Over Production	NA
O	Over-processing	NA
D	Defects	Corrected

Tangible Benefits

- Productivity has increased by 90 M.T Per Month by eliminating cutter change breakdown.
- Reduction in Conversion & overhead cost saving of 15.2 Lakhs/Annum.
- This same method is implemented in our Line no 2.

In-tangible Benefits

- Morale increased due to increase in production.
- Morale of workmen increased due to reduction in effort for changing cutters.

- Training on Updated Standard Operating Procedures are given to all the workmen for regular inspection, maintenance & to work safely.

	EVEREST INDUSTRIES LIMITED PODANUR WORKS STANDARD OPERATING PROCEDURE	ISSUE NO : 01 REV NO : 01 DATE : 04/11/2019
STANDARD OPERATING PROCEDURE FOR CUTTER CHANGE DURING MACHINE RUNNING - Updated 1) Isolate the power supply & switch OFF the cutter unit. 2) Start giving production to moulding accessories. 3) Lift up the cutter unit by using individual solenoid valve. 4) Change the cutter by using pneumatic gun. 5) Ensure the cutter unit work is complete or not. 6) After completion of cutter change, switch ON the cutter unit. 7) Down the cutter unit by using individual solenoid valve. 8) Start production to corrugated roofing.		

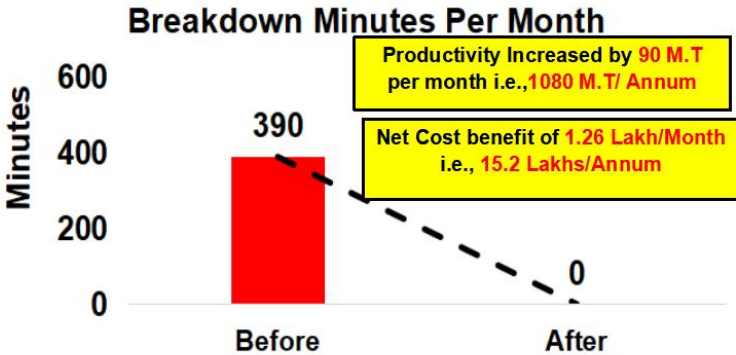
Kaizen Sustenance:

WHAT TO DO	Regular Inspection, Maintenance & safe working.
HOW TO DO	Regular Training on Standardized work instructions
FREQUENCY	Daily

Horizontal Deployment:

BW	CW	KW	LW	PW1	PW2	SW

 - Under Progress
  - Deployed

7th Edition of 3M Competition Confederation of Indian Industry INSTITUTE of QUALITY		3M KAIZEN IDEA	Activity (Please select any)	Muda ✓	Mura	Muri	everest Everest Industries Limited								
		Muda/Mura/Muri													
		Tick wherever necessary	Benefit Area	P ✓	Q	C ✓	D	S	M						
Plant : Coimbatore		Area/Line/Machine : Line No.1													
Kaizen Theme : To eliminate Machine Stoppages for Cutters & Shoes Change.		Idea (Logical correlation with root cause) : To change Cutters & Shoes while giving sheets for moulding to eliminate Machine Stoppages.													
Problem/Present Status : Loss of production.		Countermeasure (Engineering solution):				Base Data		25/10/2019							
		Before :		After :		Target		10/11/2019							
								Kaizen Start 25/10/2019							
		Fitter Changing Cutters with spanner by stopping Machine.		Fitter Changing Cutters with Pneumatic Gun		Sheets Collecting for Moulding		Kaizen Finish 4/11/2019							
		Before condition : For Cutter and Shoes change we have to stop the machine for 15 minutes Per Day i.e. 6.5 Hrs per month.		After condition : During sheet collecting for moulding, we are making each cutter ideal one by one & simultaneously changing cutters accordingly.		Team Members: Y. Kesavaraj, M. Palavesa Moorthy, N. Karthikeyan, M. Madhan Raj, G. Tulasi Rao, Jagir, Edward, Jitendar, N. Anil Kumar Reddy									
						Benefits: (P,Q,C,D,S,M) 1) Productivity has increased by 90 M.T Per Month. 2) Cutter change Breakdown has eliminated. 3) Conversion & overhead cost saving of 15.2 Lakhs/Annum.									
Analysis why - why analysis :		Results :													
Problem Phenomena : Loss of Production.		Breakdown Minutes Per Month  <table><tr><th>Minutes</th><th>Before</th><th>After</th></tr><tr><td></td><td>390</td><td>0</td></tr></table>								Minutes	Before	After		390	0
Minutes	Before									After					
	390									0					
Why 1 : Due to Machine Stoppages.															
Why 2 : Machine stoppages for Cutters & Shoes change.															
Why 3 : If Machine not stopped for Cutters & Shoe change we loss Wet Sheet production for recycling.															
Why 4 : There is NO process to collect Wet Sheets during Cutters Change without Stopping Machine.															
Root Cause : There is No process to collect Green sheets for moulding, during Cutters change.															
		Scope & Plan for Horizontal Deployment													
		SN	M/c	Target date	Responsibility	Status									
		1	Line 2	15/11/2019	Maintenance & Production Heads	Completed									

Thank You